exploratory bases, communities, and homesteads there. Because of its pioneering history, this may be a particularly natural and attractive idea in the United States.

In any case, massive alteration of the environments of other worlds can be done competently and responsibly only when we have a much better understanding of those worlds than is available today. Advocates of terraforming must first become advocates of the long-term and thorough scientific exploration of other worlds.

Perhaps when we really understand the difficulties of terraforming, the costs or the environmental penalties will prove too steep, and we will lower our sights to domed or subsurface cities or other local, closed ecological systems, greatly improved versions of Biosphere II, on other worlds. Perhaps we will abandon the dream of converting the surfaces of other worlds to something approaching the Earth's. Or perhaps there are much more elegant, cost-effective, and environmentally responsible ways of terraforming that we have not yet imagined.

But if we are seriously to pursue the matter, certain questions ought to be asked: Given that any terraforming scheme entails a balance of benefits against costs, how certain must we be that key scientific information will not thereby be destroyed before proceeding? How much understanding of the world in question do we need before planetary engineering can be relied upon to produce the desired end state? Can we guarantee a long-term human commitment to maintain and replenish an engineered world, when human political institutions are so short-lived? If a world is even conceivably inhabited—perhaps only by microorganisms—do humans have a right to alter it? What is our responsibility to preserve the worlds of the Solar System in their present wilderness states for future generations—who may contemplate uses that today we are too ignorant to foresee? These questions may perhaps be encapsulated into a final question: Can we, who have made such a mess of *this* world, be trusted with others?

It is just conceivable that some of the techniques that might eventually terraform other worlds might be applied to ameliorate the damage we have done to this one. Considering the relative